

Defending Agentic Federated Learning Systems Against Explanation Poisoning

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Abstract

Federated Learning (FL) has emerged as a foundational paradigm for privacy-preserving machine learning. Recent advances extend FL toward *agentic* systems, where autonomous clients independently curate data, optimize objectives, and adapt learning strategies. In parallel, Explainable AI (XAI) has been adopted as the primary mechanism for trust and regulatory compliance. However, recent work demonstrates that XAI itself can be adversarially manipulated, enabling *Explanation Poisoning*: attacks that preserve accuracy and plausible explanations while embedding hidden backdoors.

This work proposes a novel governance framework termed *Proof of Reasoning (PoR)*. Instead of auditing numerical model updates, PoR enforces structural alignment of causal reasoning across agents using Graph Edit Distance (GED). We show that explanation-poisoning attacks are fundamentally incompatible with causal-graph consistency, creating a mathematical contradiction that forces adversaries to either abandon stealth or abandon the attack. We provide theoretical justification, architectural design, comparative analysis against three generations of defenses, and a practical implementation roadmap.

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1 Introduction & Motivation

1.1 The Evolution: From Passive Learning to Agentic Autonomy

The field of distributed artificial intelligence is undergoing a critical transition. We are moving away from *Passive Federated Learning (FL)*—where clients function as simple gradient-computing terminals operating on static datasets—toward *Agentic Federations*. In this emerging paradigm, formally described as *Decentralized Autonomous Learning*, clients are no longer passive data holders but *Autonomous Agents*.

These agents actively curate their own training data, optimize individualized local reward functions, and interact with dynamic environments rather than fixed datasets.

The Opportunity: Agentic FL significantly reduces communication overhead and improves personalization by allowing local nodes to define and adapt their own learning curriculum.

The Risk: Autonomy introduces opacity. Unlike passive FL—where the server enforces a shared objective—an agentic node can independently alter *how* it learns. This enables a new class of attacks, including *Memory Poisoning*, where an agent subtly corrupts its long-term decision logic without triggering immediate statistical alarms.

1.2 The Core Problem: Explanation Poisoning

To govern opaque learning systems, regulators and auditors currently rely on *Explainable AI (XAI)*. The prevailing assumption is simple: *if a model can explain its decision, it can be trusted*.

Recent adversarial research (2024–2025) has fundamentally invalidated this assumption through the introduction of *Explanation Poisoning*. This attack constitutes a sophisticated *dual-objective optimization* in which the adversary trains a model to behave maliciously while producing benign-looking explanations.

Formally, the adversary solves:

$$\theta^* = \arg \min_{\theta} (\mathcal{L}_{\text{task}} + \alpha \cdot \mathcal{L}_{\text{backdoor}} + \beta \cdot \|\text{XAI}(\theta, x_{\text{bad}}) - \text{XAI}(\theta, x_{\text{good}})\|),$$

where:

- $\mathcal{L}_{\text{backdoor}}$ enforces malicious behavior (e.g., Trigger $\rightarrow$ Target),
- the XAI term enforces explanation similarity to benign samples.

Consequence: The model effectively *lies* to the human auditor. It makes decisions based on hidden triggers while generating explanations that highlight legitimate features, rendering post-hoc XAI auditing dangerously misleading.

1.3 The Trust Paradox in High-Dimensional & Medical Domains

This vulnerability is not theoretical. It creates critical failure modes in high-stakes, high-dimensional domains, which we explicitly study in this research.

Heterogeneous Healthcare (FLamby Context): In cross-silo medical federations such as FLamby, a diagnostic agent can be poisoned to systematically miss specific pathology subtypes (e.g., melanoma in Fed-ISIC2019) while highlighting irrelevant skin textures to justify benign predictions. Due to strong non-IID data distributions across hospitals, statistical defenses struggle to differentiate malicious logic from natural heterogeneity.

Biometrics & Attribute Logic (CelebA Context): In facial analysis systems, an agent can be manipulated to rely on spurious correlations—such as using *Heavy Makeup* as the sole predictor of *Gender*—while masking this bias from auditors. This creates the classic “*Right Answer, Wrong Reason*” paradox.

The Trust Paradox: As systems become more explainable, attackers gain more surface area to manipulate trust. Current regulations mandate explainability, but do not mandate *explanation integrity*. This constitutes a fundamental regulatory and technical blind spot.

1.4 Research Objectives & Contributions

This project aims to close the emerging *Trust Gap* in Agentic Federated Learning by moving beyond result-based auditing. Our core objectives are:

- **Formalizing Logic-Based Governance:** Redefining trust from Euclidean proximity in weight space to *topological consistency* in causal logic space.
- **Defeating Explanation Poisoning:** Designing a system in which it is mathematically impossible for an agent to misrepresent its reasoning without detection.
- **Enabling Scalable Verification:** Demonstrating that Graph Edit Distance (GED) can be efficiently approximated and deployed as a real-time governance mechanism in modern federated systems, validated across both visual attributes and real-world medical silos.

2 Literature Review & Research Gap Analysis

The landscape of Trusted AI has evolved significantly between 2024 and 2026. While Federated Learning (FL) has matured as a deployment paradigm, adversarial objectives have shifted from overt accuracy degradation to subtle manipulation of trust mechanisms. This section reviews the state-of-the-art across four critical dimensions and identifies the precise research gap addressed by Proof of Reasoning (PoR).

2.1 The Evolving Threat: From Poisoning to “Explanation Scaffolding”

Early adversarial attacks (2017–2020) primarily focused on degrading model accuracy through techniques such as label flipping and data poisoning. Recent literature, however, highlights a shift toward *stealth attacks*, where predictive performance is preserved while trust and interpretability are silently compromised.

Explanation Scaffolding (2024): Liyanage et al. introduced the concept of *Scaffolding Attacks* in network intrusion detection systems. They demonstrated that an adversary can embed a persistent bias (e.g., systematically ignoring a specific IP range) while simultaneously manipulating the model to conceal this bias from post-hoc explainability tools such as SHAP.

Fooling SHAP (2024): Subsequent work showed that “Output Shuffling Attacks” can wrap a model with a surrogate function. This allows the model to act on protected attributes (e.g., race) while the SHAP explainer reports near-zero attribution for those features.

Healthcare Vulnerabilities (2026): A comprehensive clinical AI review found federated healthcare systems to be particularly vulnerable. As few as 100–500 poisoned samples were sufficient to implant a backdoor in a 13B-parameter medical model, without noticeable performance degradation.

Our Critique: These works convincingly demonstrate the existence and severity of explanation-level attacks. However, they stop short of defining a principled defense. The common recommendation of “better auditing” lacks a concrete mathematical formulation of what should be audited and how.

2.2 Defenses in Federated Learning: The “Statistical” Stalemate

Most federated defenses continue to rely on statistical analysis of model updates in weight space.

FedREDefense (ICML 2024): Introduced reconstruction error via autoencoders to detect anomalous updates. Updates that cannot be efficiently reconstructed are rejected.

Failure Mode: This approach is effective against noisy or unstructured attacks. Explanation Poisoning attacks, however, are deliberately clean and structured, allowing them to pass reconstruction-based filters.

ESFL (IEEE TDC 2025): Proposed Efficient and Secure Federated Learning using adaptive local differential privacy (LDP) and compressive sensing.

Failure Mode: While LDP improves privacy, the injected noise paradoxically aids attackers by obscuring malicious triggers. The fundamental safety–privacy trade-off remains unresolved.

Triple-Phase Sanitization (MDPI 2026): A recent NLP-focused defense using text readability and word-frequency statistics to detect poisoned samples.

Limitation: This method is domain-specific and does not generalize to vision-based or medical federated systems such as CelebA or FLamby, which are central to agentic FL.

2.3 Causal Discovery in Federated Systems

Federated Causal Discovery (FCD) has emerged as a promising research direction, but its focus has been performance rather than security.

FedECD (NeurIPS 2024): Proposed learning global causal graphs from clients with limited local data using bootstrapped causal skeleton refinement.

FedCausal (AAAI 2025): Introduced a global optimization framework to learn a unified DAG under non-IID data distributions.

Critical Gap: Both approaches use causal graphs as a tool for feature engineering and accuracy improvement. They implicitly assume all clients are honest contributors seeking the true causal structure.

Our Insight: No existing work considers adversarial clients intentionally submitting *fake causal graphs*. The use of causality as a *security and auditing mechanism* remains entirely unexplored. This is the precise gap addressed by PoR.

2.4 Graph-Based Metrics: The Enabler

A key challenge for any graph-based defense is computational feasibility. Historically, Graph Edit Distance (GED) computation is NP-hard.

GRAPHEDX (NeurIPS 2024): Introduced Neural Set Divergence, a GNN-based approximation of GED that estimates graph distances in milliseconds rather than exponential time.

Strategic Relevance: This breakthrough makes real-time, graph-based governance feasible. PoR directly adopts this approximation to implement a practical server-side logic validator.

2.5 Summary of the Research Gap

Domain	Current State (2025–2026)	Missing Piece (PoR)
Attack	Explanation Poisoning undetectable by SHAP	Validation of explanations themselves
Defense	Weight statistics / Differential Privacy	Structural logic-based defense
Causality	Used for accuracy (FedECD, FedCausal)	Causality for auditing and trust
Mathematics	Neural GED solved but unused in FL	Neural GED as a security gatekeeper

Table 1: Summary of the research gap addressed by Proof of Reasoning (PoR).

3 Proposed System Design & Architecture

This section outlines the architectural blueprint of the **Proof of Reasoning (PoR)** framework. We adopt a *Hierarchical Agentic Architecture* in which decentralized nodes operate as *Deliberative Agents* capable of internal world-modeling, while the central server functions as a *Governance Orchestrator*. The design explicitly separates performance optimization from logical governance.

3.1 Architectural Overview

The system extends the standard client–server federated learning paradigm with a specialized *Governance Layer*.

- **Local Agent (The Learner):** Optimizes the local task (e.g., facial attribute classification on CelebA) while simultaneously extracting a *Logic Signature*.
- **Governance Layer (The Gatekeeper):** A server-side module that intercepts client updates. Instead of blindly accepting parameters, it validates the structure of the agent’s reasoning using graph-based analysis.
- **Aggregation Layer:** Performs a *Dual-Stream Update*, aggregating model weights for performance and causal graphs for logic consensus.

3.2 Client-Side Architecture (The “Deliberative Agent”)

Each client is modeled as an autonomous deliberative agent composed of three distinct modules. Unlike standard federated learning clients that only compute gradients, PoR agents explicitly compute and transmit causal explanations.

3.2.1 The Perception Module (Data Ingestion)

Function: Ingests the local private dataset $\mathcal{D}_k$.

Novelty: Implements *Causal Disentanglement*. Instead of feeding raw inputs directly into the model, features are separated into:

- **Invariant Mechanisms:** Stable causal factors (e.g., eye shape),
- **Spurious Correlations:** Noise or potential trigger features.

This reduces the model’s reliance on shortcuts and prepares the representation for causal analysis.

3.2.2 The Cognitive Module (Reasoning Extraction)

This module is the core innovation of PoR. As the agent trains its local model θ_k , it simultaneously extracts the underlying decision logic.

Algorithmic Principle: A gradient-based causal discovery method adapted for neural networks.

Process:

1. Latent layers of the neural network are treated as variables in a Structural Equation Model (SEM).
2. Gradients and activation dependencies are analyzed to infer directed causal influence.
3. An adjacency matrix A_k is constructed, where:

$$A_{ij} = 1 \implies \text{Feature } i \text{ causally influences Feature } j.$$

Output: A directed acyclic graph (DAG) G_k that serves as the agent’s *Logic Signature*.

3.2.3 The Action Module (The Update Payload)

Rather than transmitting a single parameter vector, each agent sends a structured update:

$$\text{Payload}_k = \{\Delta\theta_k, G_k\},$$

where:

- $\Delta\theta_k$ is the weight update (performance signal),
- G_k is the causal graph (reasoning signal).

3.3 Server-Side Architecture (The “Governance Layer”)

The server acts as a logical gatekeeper. It does not blindly aggregate updates, but first validates the reasoning structure using graph-based learning.

3.3.1 The Logic Validator (Neural GED Estimator)

Exact computation of Graph Edit Distance (GED) is NP-hard. To enable real-time governance, we use a neural approximation.

Mechanism:

- The server maintains a pre-trained *Siamese Graph Neural Network (GNN)*.
- Inputs are the client graph G_k and the global consensus graph G_{global} .

Operation:

$$\text{Score}_k = \|\text{GNN}(G_k) - \text{GNN}(G_{\text{global}})\|_2.$$

Decision Rule:

If $\text{Score}_k > \tau \implies$ Update is Logically Anomalous and Rejected.

3.3.2 The Dual Aggregator

After filtering out anomalous agents, the server performs two parallel aggregations.

Weight Aggregation: Uses FedProx to handle non-IID data distributions:

$$\theta_{t+1} = \sum_{k \in \text{Accepted}} \frac{n_k}{N} \theta_k.$$

Logic Aggregation: Uses barycenter graph averaging to update the global consensus graph G_{global} .

- Edges appearing in more than 50% of accepted agents are retained.

Significance: This allows the federation to gradually incorporate new, valid causal relationships over time, provided they are supported by a majority of honest agents.

3.4 The Security Protocol (The Handshake)

The system operates in synchronized communication rounds:

1. **Broadcast:** Server sends $(\theta_t, G_{\text{global}})$ to all agents.
2. **Local Optimization:** Each agent updates θ_k to minimize task loss and a logic consistency term:

$$\mathcal{L}_{\text{local}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \|G_{\text{local}} - G_{\text{global}}\|.$$

3. **Audit:** Agent uploads $\{\Delta\theta_k, G_{\text{local}}\}$.
4. **Verification:** Server evaluates the neural GED score.
5. **Decision:**
 - Honest Agent: $\text{GED} \approx 0$.
 - Adversary: $\text{GED} \gg 0$.
6. **Rejection:** Logically anomalous agents are excluded from aggregation.

3.5 Component Summary Table

Component	Standard (SOTA 2023)	PoR Implementation
Agent Model	ResNet / Transformer	Causal-Informed Neural Network
Privacy	Differential Privacy (Noise)	Structural Masking (Graph-Only)
Aggregation	Weighted Average	Dual-Stream (Weights + Topology)
Audit Metric	Cosine Similarity	Neural Graph Edit Distance

Table 2: Comparison of standard federated systems and the PoR architecture.

4 Mathematical Framework & Comparative Derivation

This section provides the mathematical foundation of our proposal. We frame Explanation Poisoning as a bilevel optimization problem, rigorously showing why adversaries succeed against Euclidean defenses and how a topological reasoning-based defense introduces a fundamental contradiction that defeats them.

4.1 Problem Formulation: The Adversarial Objective

Let $\mathcal{D} = \{(x, y)\}$ denote the local dataset. Let θ be the model parameters and $f_\theta(x)$ the prediction function. Let $E_\theta(x)$ be an explanation function (e.g., gradient-based saliency) that returns a feature-importance map.

A sophisticated Explanation Poisoning attacker seeks to solve the following dual-objective optimization problem:

$$\min_{\theta^*} \left(\underbrace{\mathcal{L}_{CE}(f_{\theta^*}(x_{\text{trigger}}), y_{\text{target}})}_{\text{Backdoor Success}} + \beta \cdot \underbrace{\|E_{\theta^*}(x_{\text{trigger}}) - E_{\theta_{\text{clean}}}(x_{\text{clean}})\|_2^2}_{\text{Explanation Camouflage}} \right).$$

Constraint 1 (Stealth):

$$\|\theta^* - \theta_{\text{global}}\|_2 < \epsilon.$$

Key Observation: Deep neural networks are highly non-convex and over-parameterized. As a result, an adversary can find a solution θ^* that simultaneously satisfies the stealth constraint and successfully poisons both the prediction and its explanation.

4.2 Failure Proof of Existing Defenses (The Euclidean Flaw)

Standard federated defenses (e.g., Krum, FedProx) rely on Euclidean distance in parameter space. A server accepts a client update $\Delta\theta$ if:

$$\text{Accept}(\Delta\theta) \iff \|\Delta\theta - \Delta\theta_{\text{avg}}\|_2 \leq \delta.$$

Derivation of Failure:

Let $\mathcal{S}_{\text{safe}}$ denote the hypersphere of radius δ centered at the global model. Let $\mathcal{S}_{\text{attack}}$ denote the manifold of parameters that successfully encode the backdoor.

Due to the curse of dimensionality in modern deep networks (with millions of parameters), the intersection

$$\mathcal{S}_{\text{safe}} \cap \mathcal{S}_{\text{attack}} \neq \emptyset.$$

Conclusion: An attacker using Projected Gradient Descent (PGD) can explicitly project malicious updates onto $\mathcal{S}_{\text{safe}}$. The update passes all Euclidean checks, yet the model is poisoned. The mathematics validates the update, but the security objective fails.

4.3 The Proposed Metric: Topological Graph Edit Distance (GED)

We abandon Euclidean parameter space in favor of *Topological Causal Space*. Instead of comparing weights, we compare *logic graphs*.

Let $G = (V, E)$ be a causal directed acyclic graph (DAG), where:

- V are latent feature nodes,
- E are directed causal edges.

The Graph Edit Distance (GED) between a client graph G_c and the global consensus graph G_g is defined as:

$$GED(G_c, G_g) = \min_{(op_1, \dots, op_k) \in \Gamma(G_c, G_g)} \sum_{i=1}^k \text{Cost}(op_i),$$

where $\Gamma(G_c, G_g)$ is the set of valid edit-operation sequences transforming G_c into G_g .

Allowed Operations:

- **Vertex substitution:** $v \rightarrow u$ (changing a feature node),
- **Edge insertion/deletion:** e^+, e^- (adding or removing a causal link).

Novel Cost Function: Edges connected to *Protected Features* (e.g., clinically critical symptoms or sensitive facial attributes) incur a higher penalty:

$$\text{Cost}(e_{\text{critical}}^-) \gg \text{Cost}(e_{\text{noise}}^-).$$

4.4 The New Governance Objective

We propose a joint optimization objective that every federated client must satisfy:

$$\mathcal{L}_{\text{PoR}}(\theta_k) = \mathcal{L}_{\text{Task}}(\theta_k) + \lambda \cdot \underbrace{GED(\text{Extract}(\theta_k), G_{\text{global}})}_{\text{Structural Regularizer}}.$$

Here:

- $\text{Extract}(\cdot)$ is a differentiable causal discovery operator (e.g., via Gumbel-Softmax),
- λ is the *Logic Compliance Coefficient*.

This forces each client to optimize not only for task accuracy, but also for logical conformity with the federation.

4.5 Theoretical Proof of Defense (The Contradiction)

Theorem: An adversary cannot simultaneously satisfy the Explanation Poisoning attack objective and the PoR defense objective.

Proof (by Contradiction):

Assume an adversary successfully injects a backdoor (e.g., $Trigger \rightarrow Output$) while maintaining a low GED score.

Causal Requirement: For the backdoor to function reliably, there must exist a causal path P_{attack} in the attack graph G_{attack} connecting node V_{trigger} to node V_{output} .

Consensus Conflict: The global consensus graph G_{global} , derived from honest agents, does not contain this path, as benign data does not support such causality.

Distance Implication: To transform G_{attack} into G_{global} , the adversary must delete P_{attack} , yielding:

$$GED(G_{\text{attack}}, G_{\text{global}}) \geq \text{Cost}(P_{\text{attack}}) > 0.$$

Catch-22:

- Removing the path minimizes GED but destroys the backdoor (attack fails).
- Keeping the path preserves the backdoor but yields a high GED (defense triggers).

Conclusion: Unlike Euclidean weight space—where attacks can hide in high-dimensional subspaces—causal logic space exposes the attack mechanism itself. The adversarial and defensive objectives are mutually exclusive.

5 Comparative Analysis & Performance Benchmarking

To demonstrate the necessity of the **Proof of Reasoning (PoR)** framework, we benchmark it against three distinct generations of Federated Learning defense mechanisms. We analyze why each generation fails to detect *Explanation Poisoning* and how our approach fills these critical gaps.

5.1 The Evolution of Defenses (Generational Analysis)

5.1.1 Generation 1: Statistical Outlier Detection (2017–2020)

Representative Algorithms: Krum (NeurIPS 2017), Trimmed Mean, Median.

Mechanism: These algorithms assume that malicious updates are *statistical outliers* in Euclidean space. They calculate the spatial center of all client updates and reject those that are furthest away.

Failure Mode (The “Hidden Backdoor”):

- **Mathematical Failure:** An adversary using Projected Gradient Descent (PGD) can constrain a poisoned update w_{bad} such that

$$\|w_{\text{bad}} - w_{\text{avg}}\| < \epsilon.$$

- **Result:** The poisoned model becomes statistically indistinguishable from a benign model. Krum therefore selects it as “safe.”

PoR Advantage: PoR does not evaluate the position of updates in vector space. Instead, it analyzes the *structure of the reasoning graph*. A hidden backdoor requires a divergent logical path (e.g., *Trigger* $\rightarrow$ *Output*), which alters graph topology even when weight distance remains small.

5.1.2 Generation 2: Vector & Similarity Analysis (2021–2023)

Representative Algorithms: FoolsGold (USENIX 2020), FLTrust (NDSS 2021), CONTRA (ESORICS 2021).

Mechanism:

- **FoolsGold** detects Sybil attacks by identifying updates that are overly similar using cosine similarity.
- **FLTrust** validates client updates using a small server-side *Root Dataset*.

Failure Mode (Privacy & Non-IID Data):

- **FoolsGold Failure:** In agentic federated settings (e.g., FLamby), honest agents often learn similar features, resulting in high cosine similarity. FoolsGold therefore produces false positives by flagging honest, high-performing agents as Sybils.
- **FLTrust Failure:** FLTrust violates privacy by requiring server-side data. It also assumes that the Root Dataset covers all relevant edge cases. Attacks targeting rare conditions (e.g., a rare lesion in ISIC2019) bypass this defense.

PoR Advantage: PoR is *data-free*. It compares logic graphs rather than raw data samples, eliminating the need for a Root Dataset and preserving privacy.

5.1.3 Generation 3: Advanced Reconstruction & Clustering (2024–2025)

Representative Algorithms: FedREDefense (ICML 2024), FLAME (USENIX 2022/2024), Dim-Krum.

Mechanism: These defenses rely on reconstruction error (via autoencoders) or clustering clients based on update patterns.

Failure Mode (Explanation Poisoning):

- **Core Gap:** These methods detect anomalous *weights*, not anomalous *reasoning*.
- **Result:** Explanation Poisoning attacks explicitly optimize for plausible explanations. Heatmaps appear normal, and reconstruction error remains low because the backbone network is largely frozen.

PoR Advantage: PoR explicitly audits the *causal DAG*. Even if explanation pixels are falsified, internal causal dependencies cannot be altered without degrading model accuracy.

5.2 Detailed Comparison Matrix

Feature / Metric	Gen 1	Gen 2	Gen 3	PoR
Primary Metric	Euclidean Distance	Cosine / Root Data	Reconstruction / Clusters	Graph Edit Distance
Trust Assumption	Honest Majority	Server Data Exists	Weights Look Different	Logic Must Align
Privacy Risk	Low	High	Low	Zero
Defense: Backdoors	Fails	Fails	Moderate	Robust
Defense: Exp. Poisoning	Fails Completely	Fails Completely	Fails	Robust
Computation Cost	Very Low ($O(d)$)	Low ($O(d)$)	High (Autoencoders)	High (GNNs)
Non-IID Robustness	Poor	Poor	Moderate	High

Table 3: Comparison of Federated Learning defenses against Proof of Reasoning (PoR).

5.3 Theoretical Performance Derivation

Let an adversary optimize a poisoned model θ' as:

$$\theta' = \arg \min_{\theta} (\mathcal{L}_{\text{class}} + \alpha \cdot \mathcal{L}_{\text{backdoor}} + \beta \cdot \mathcal{L}_{\text{explanation_fake}}).$$

5.3.1 Why Distance-Based Defenses Fail

The adversary imposes a constraint:

$$\|\theta' - \theta_{\text{global}}\| < \epsilon.$$

Due to over-parameterization, a solution θ' exists in a subspace arbitrarily close to θ_{global} . Distance-based defenses observe

$$\text{Dist}(\theta') \approx 0$$

and accept the update.

5.3.2 Why PoR Succeeds

PoR evaluates the discrete graph structure:

$$G(\theta') = \text{CausalDiscovery}(\theta').$$

To activate a backdoor, the causal path

$$\text{Trigger} \rightarrow \text{Hidden Layer} \rightarrow \text{Output}$$

must exist.

Adversary’s Dilemma:

- Removing this edge disables the backdoor.
- Keeping the edge increases the Graph Edit Distance beyond the acceptance threshold.

Thus, the adversary cannot satisfy both the attack objective and the PoR defense simultaneously.

5.4 Known Limitations (Self-Critique)

Frozen Graph Problem: An overly rigid global consensus graph G_{global} may reject honest agents that discover new, valid causal relationships.

Mitigation: We introduce a *decay factor* that gradually weakens unused edges, allowing the system to incorporate new logic over time.

Computational Overhead: Exact Graph Edit Distance computation is NP-hard and infeasible for large models.

Mitigation: We approximate GED using neural graph matching with Siamese GNNs, reducing complexity from exponential to polynomial time.

6 Conclusion & Strategic Impact

6.1 Summary of Contributions

This project identifies and addresses a critical vulnerability in the emerging field of *Agentic Federated Learning*. As the industry moves toward autonomous, decentralized AI systems, our research demonstrates that existing safety protocols—particularly those relying on Euclidean distance metrics and post-hoc XAI auditing—are mathematically insufficient against *Explanation Poisoning* attacks.

Our primary contribution is the introduction of **Proof of Reasoning (PoR)**, a novel governance protocol that fundamentally redefines the notion of *trust* in federated systems. By elevating *Causal Graph Edit Distance (GED)* to a first-class security metric, this work delivers three key technical breakthroughs:

- **Logic-Based Auditing:** We shift the defense surface from *parameter space*, where attacks can be masked by over-parameterization, to *topological space*, where attacks necessarily disrupt causal structure.
- **The “Catch-22” Defense:** We theoretically prove (Section 4.5) that minimizing GED forces an adversary into a mathematical contradiction: they cannot preserve a functional backdoor while simultaneously maintaining a graph structure that conceals it.

6.2 The Paradigm Shift: From “Black Box” to “Glass Box” Governance

For over a decade, Federated Learning has operated under a *Black Box* trust model, implicitly trusting the statistical average of client updates. Our research argues that this assumption no longer holds in an era of autonomous, agentic AI.

We advocate a transition to *Glass Box Governance*, where trust is no longer passive but actively enforced. In this paradigm, agents must be transparent not only about their outputs, but about their internal reasoning processes. PoR serves as a blueprint for this shift: rather than merely filtering malicious updates, it enforces a protocol in which agents are required to *show their work*.

6.3 Broader Impact & Future Relevance

The implications of this research extend beyond academic benchmarks and into real-world deployment:

Regulatory Compliance: With emerging regulatory frameworks such as the EU AI Act mandating transparency and human oversight for high-risk AI systems, PoR provides a concrete technical mechanism for enforcing these requirements in decentralized environments where no single authority has access to the full dataset.

Healthcare Validation: Through validation on the FLamby benchmark (Fed-ISIC2019), we demonstrate that PoR is robust not only on curated academic datasets but also in highly heterogeneous, cross-silo medical settings. This establishes confidence that deployed medical AI systems are reasoning causally rather than relying on spurious correlations—a prerequisite for safe clinical adoption.

Final Statement

Adversarial Machine Learning has traditionally been a reactive game of cat-and-mouse: attackers exploit a loophole, and defenders patch it. This project aims to break that cycle. By anchoring trust in the *causal logic* of decision-making rather than in opaque numerical parameters, we propose a federated learning ecosystem that is resilient not only to current threats, but also to the deceptive, autonomous attacks of the future.

7 Signatures

Project Guide:

Students